Physics of Induction Cooking - Answer Sheet

C.1. Characterisation of the induction coil (4.5 pts)

1.1 (0.4 pt)

$R_{cable} =$

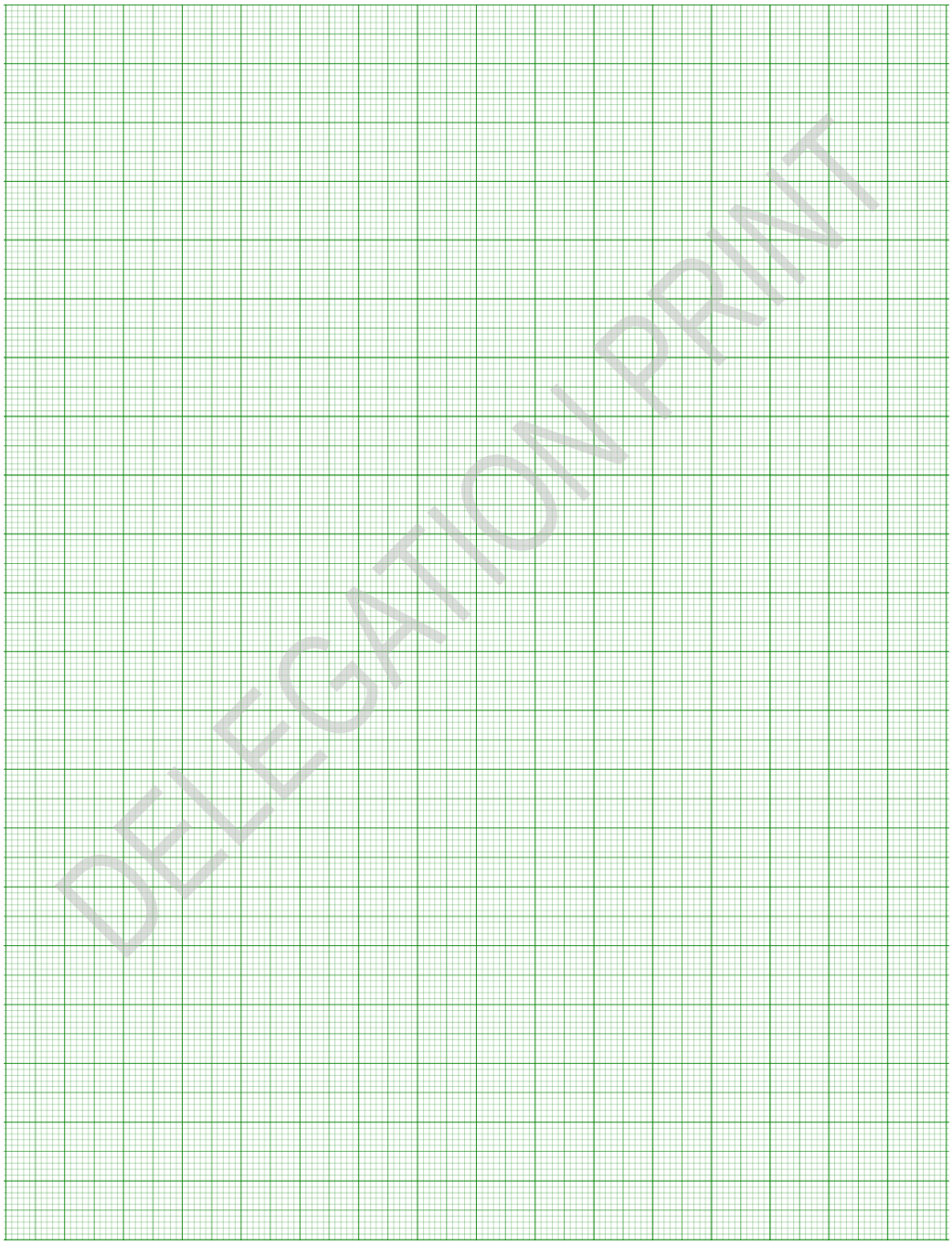
1.2 (1.2 pt)

DELEGATION PRINT

1.2 (cont.)

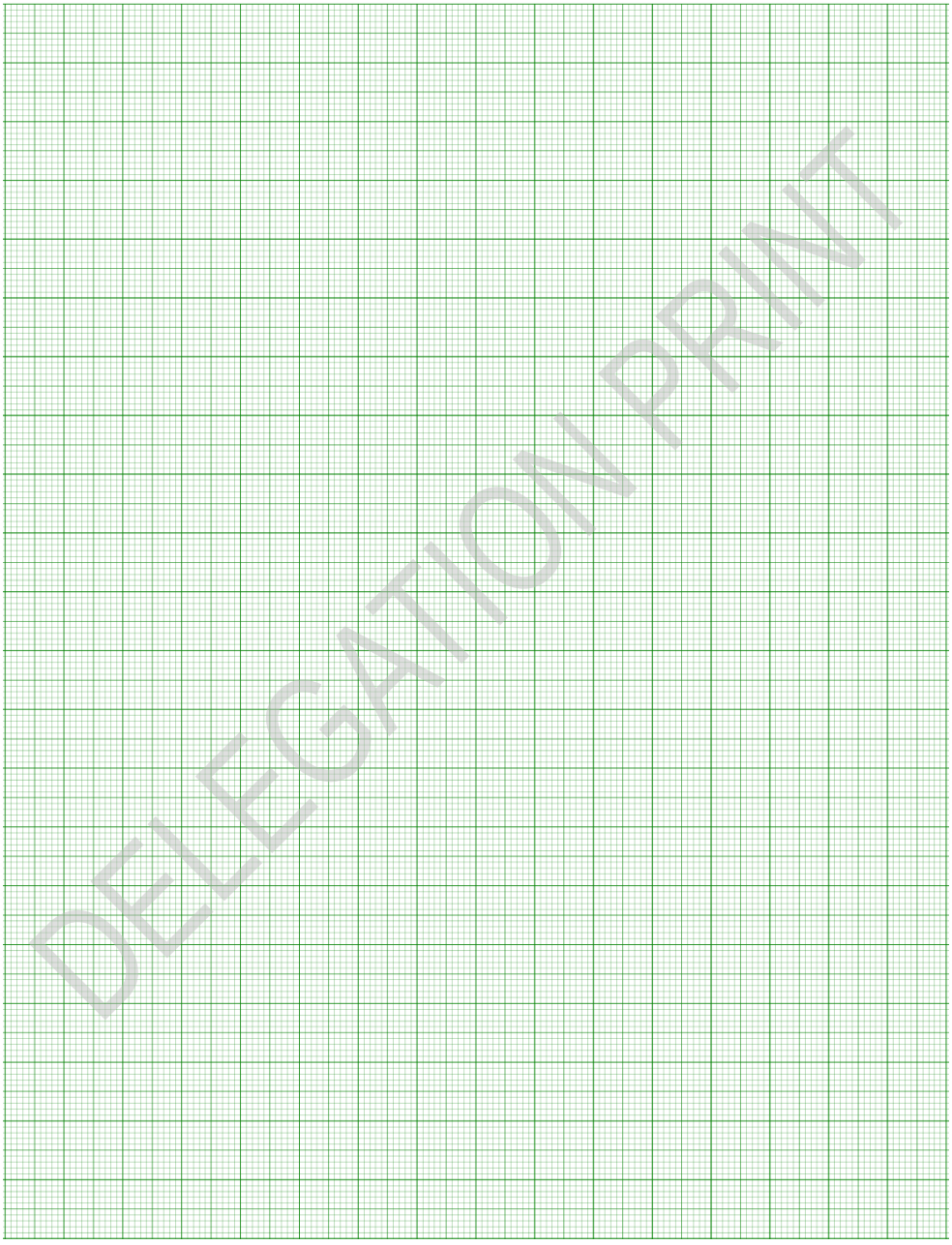
DELEGATION PRINT

1.2 (cont.)



DELEGATION PRINT

1.2 (cont.)



DELEGATION PRINT

$(C = 470\,\mathrm{nF})$, $L =$

$(C = 2200\,\mu\mathrm{F})$, $L =$

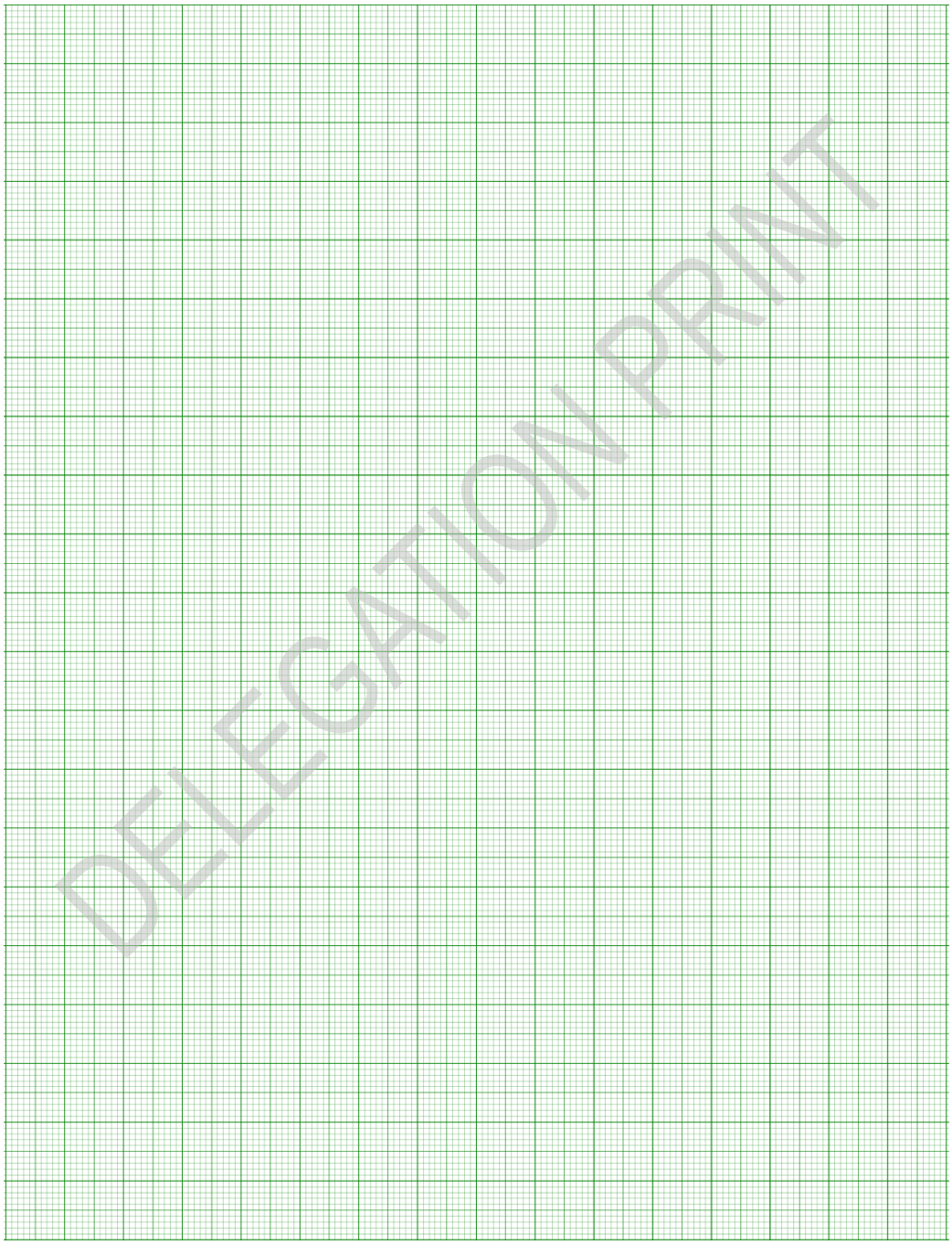
1.3 (0.5 pt)

DELEGATION PRINT

1.4 (1.4 pt)

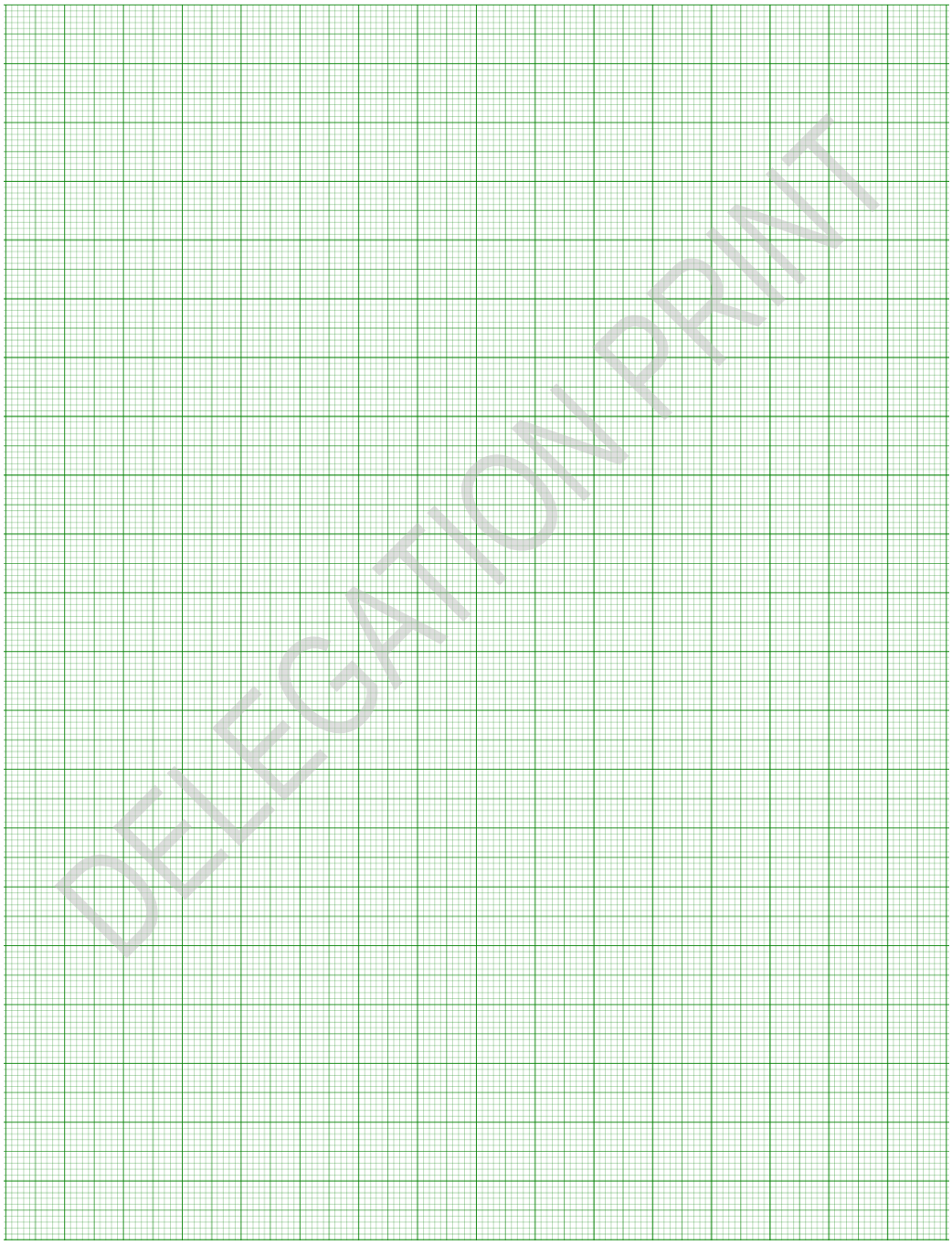
DELEGATION PRINT

1.4 (cont.)



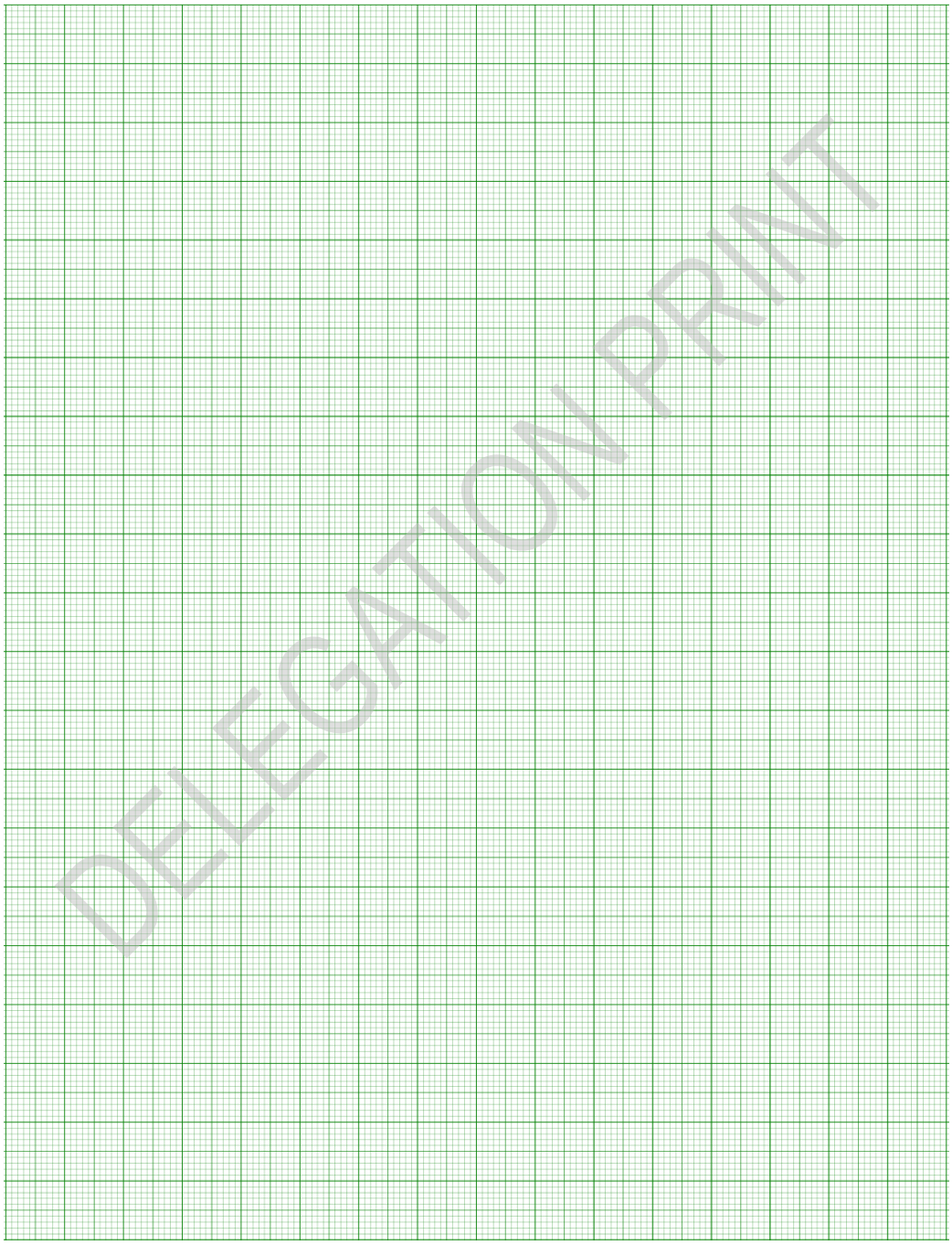
DELEGATION PRINT

1.4 (cont.)



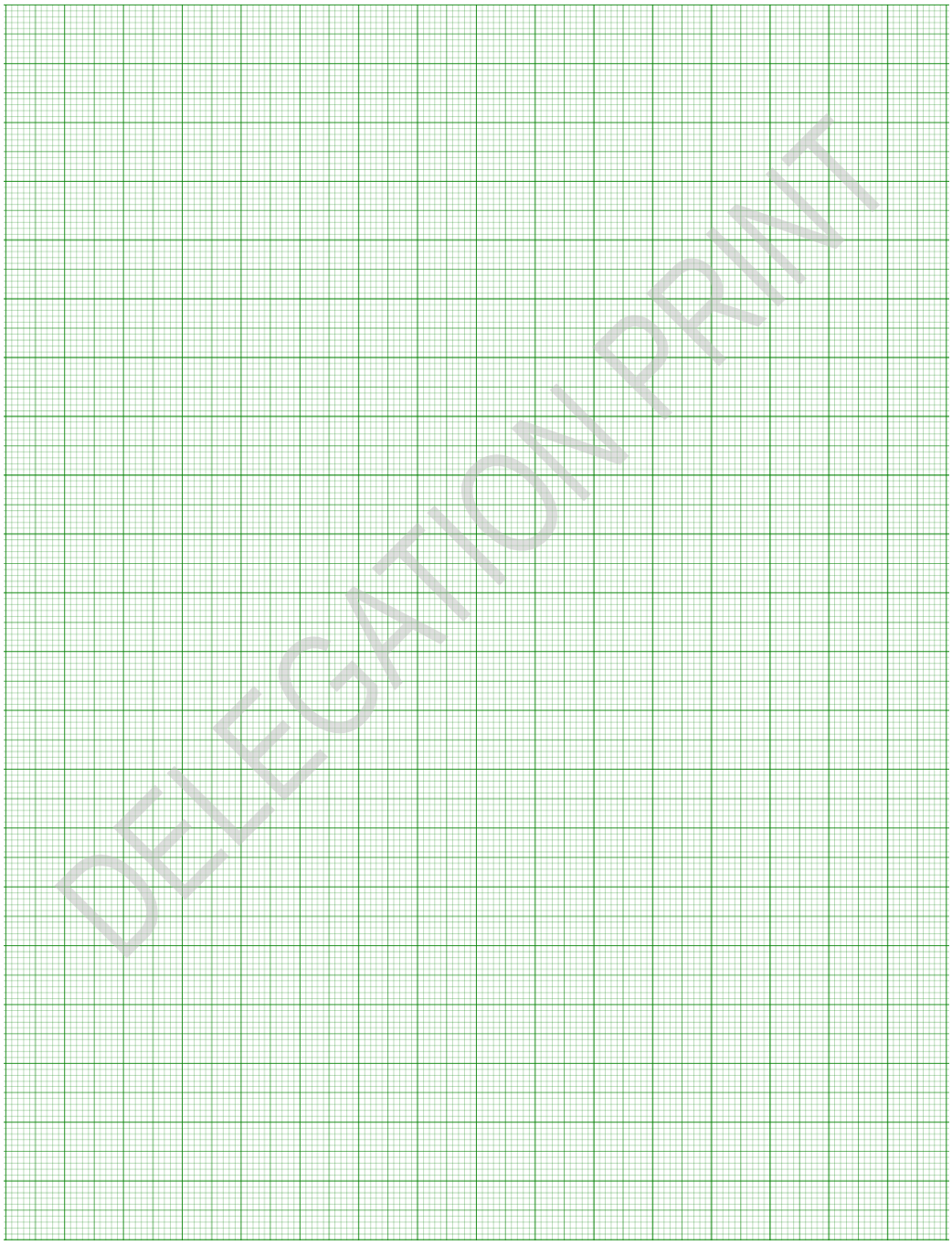
DELEGATION PRINT

1.4 (cont.)



DELEGATION PRINT

1.4 (cont.)



DELEGATION PRINT

1.5 (1 pt)

$L(\text{average}) =$

$R_L(\text{average}) =$

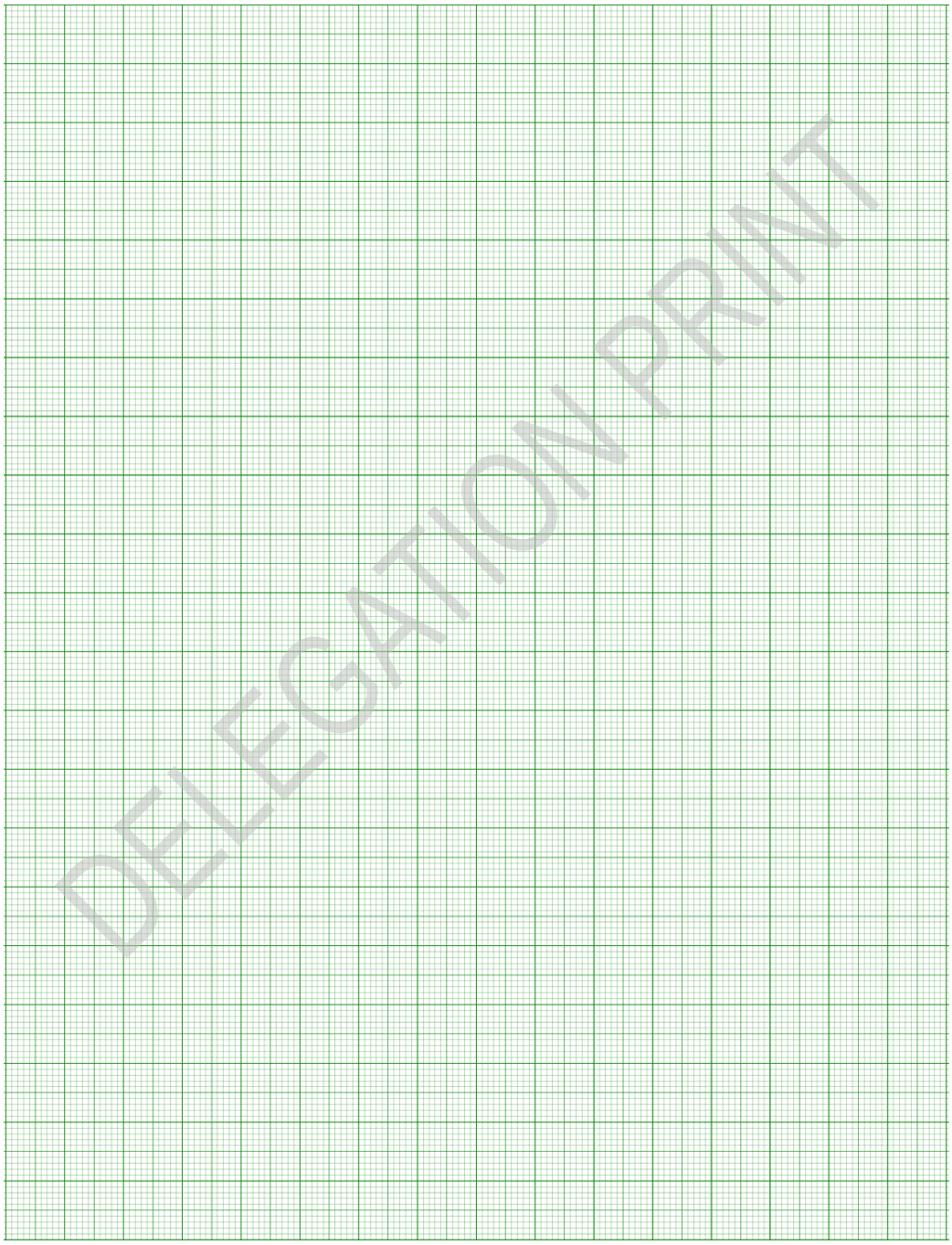
C.2. Mutual Induction and Skin Depth (8.1 pts)

2.1 (0.4 pt)

2.2 (1 pt)

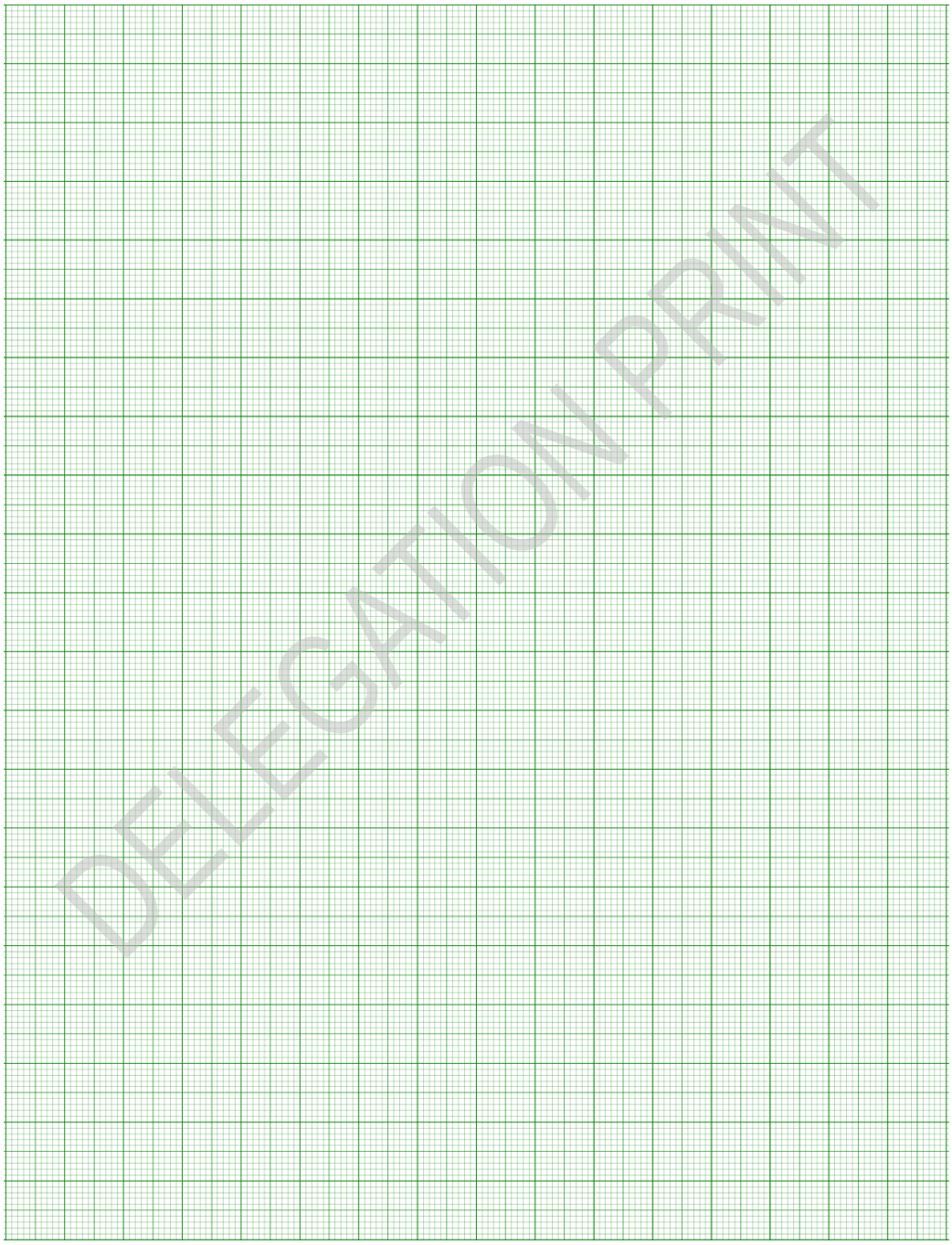
DELEGATION PRINT

2.2 (cont.)



DELEGATION PRINT

2.2 (cont.)



DELEGATION PRINT

2.3 (0.4 pt)

$M =$ $M =$

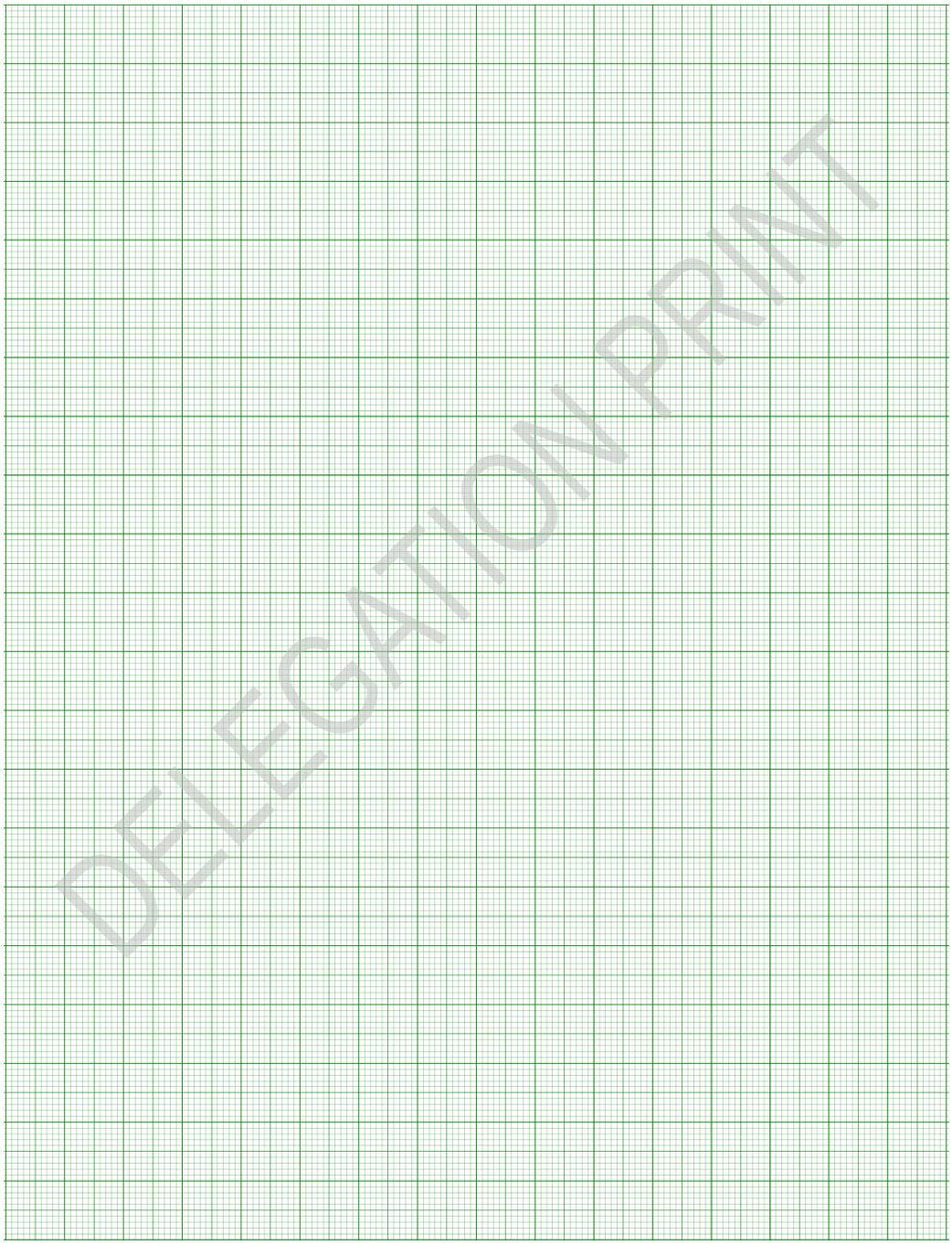
2.4 (5.5 pt)

DELEGATION PRINT

2.4 (cont.)

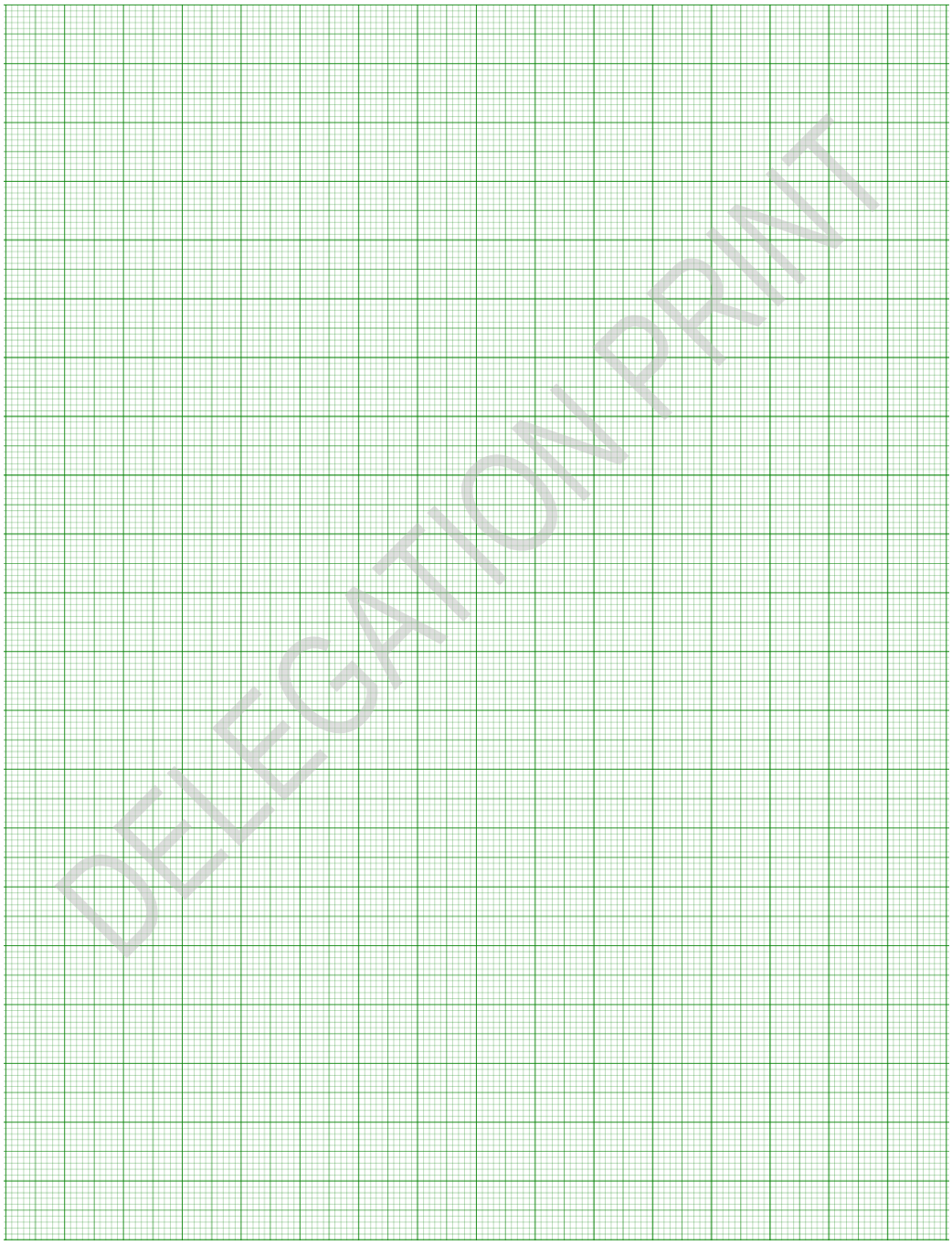
DELEGATION PRINT

2.4 (cont.)



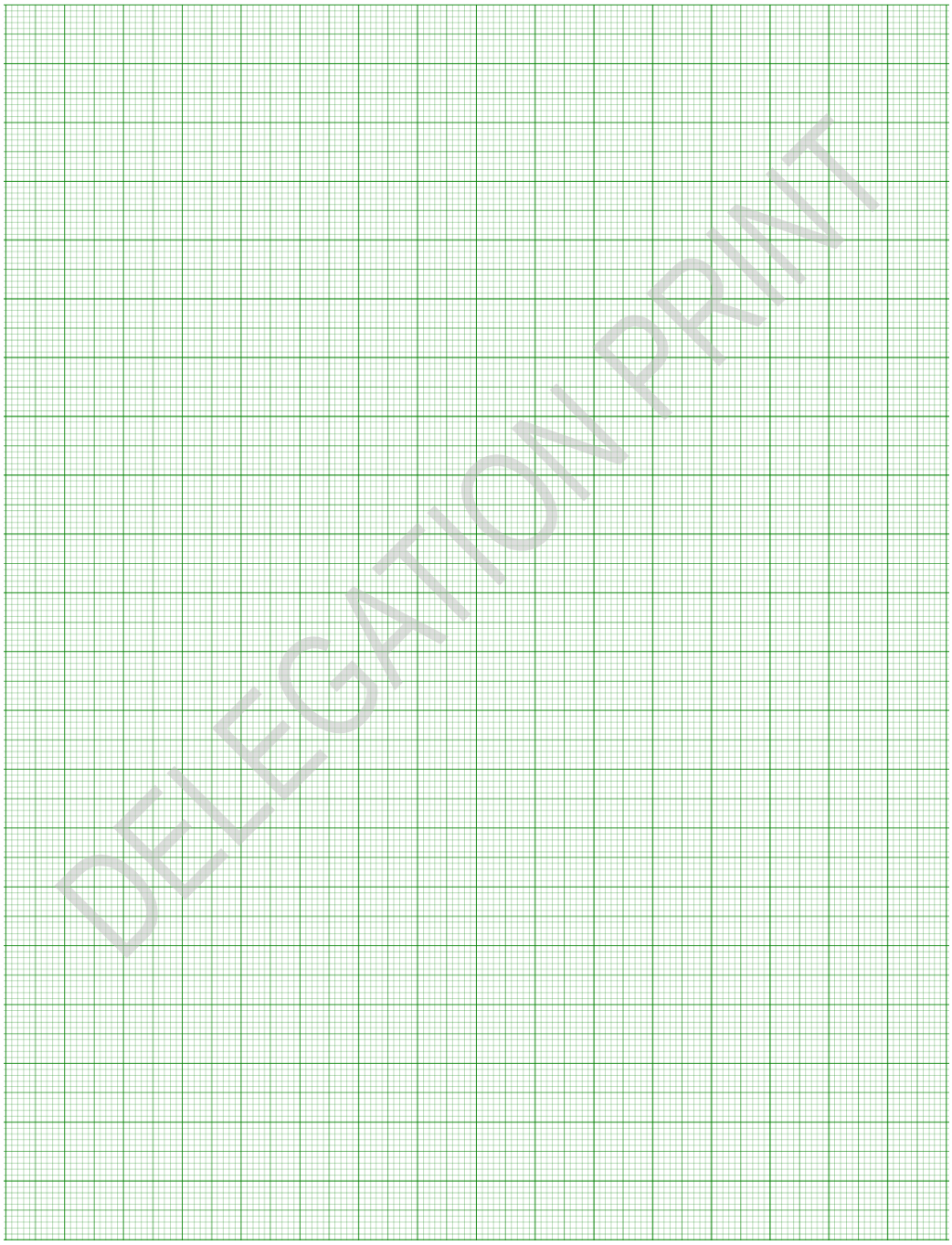
DELEGATION PRINT

2.4 (cont.)



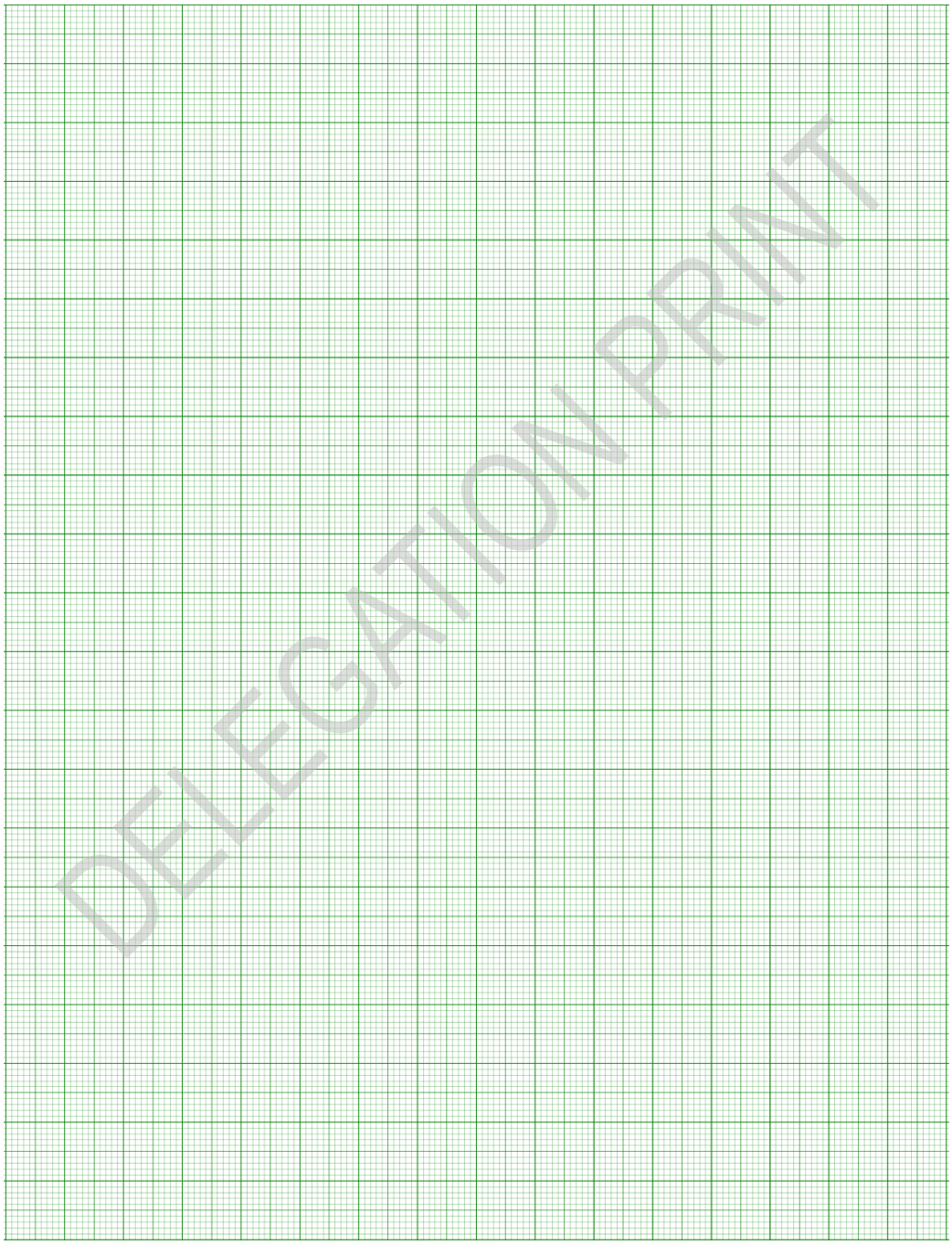
DELEGATION PRINT

2.4 (cont.)



DELEGATION PRINT

2.4 (cont.)



DELEGATION PRINT

2.4 (cont.)

In the boxes below, tick one metal that does not yield good data due to extreme value of skin depth.
☐ SS410 ☐ SS304 ☐ Cu ☐ Al

For the other three metals, state their *n* values below, with appropriate subscripts.

n = n = n =

2.5 (0.2 pt)

$m =$

2.6 (0.6 pt)

State the values of σ of the three metals that do yield good data, with appropriate subscripts.

σ = σ = σ =

C.3. : Specific heat capacity and the effective load resistance (7.4 pts)

3.1 (0.2 pt)

DELEGATION PRINT

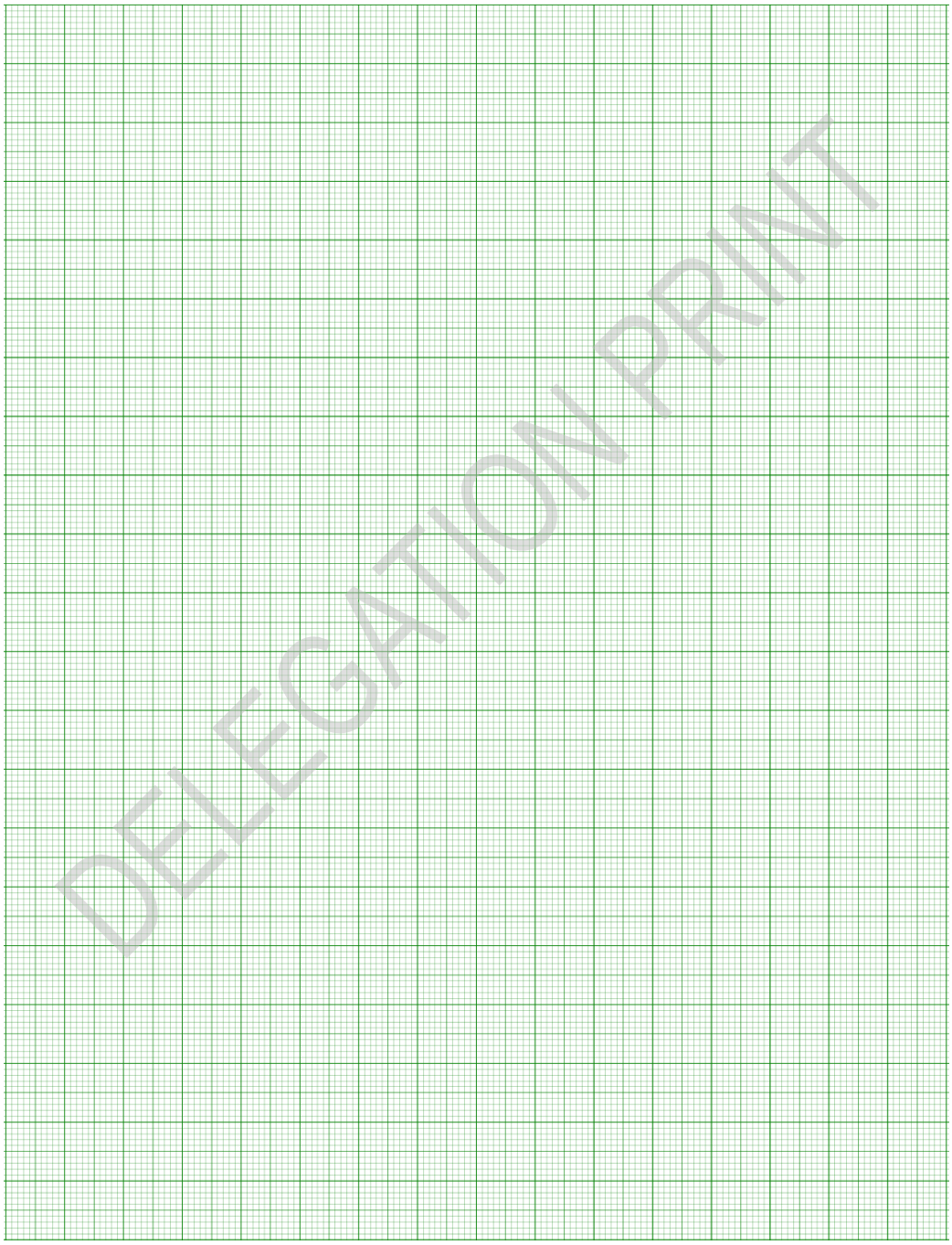
3.2 (0.5 pt)

DELEGATION PRINT

3.3 (1.5 pt)

DELEGATION PRINT

3.3 (cont.)



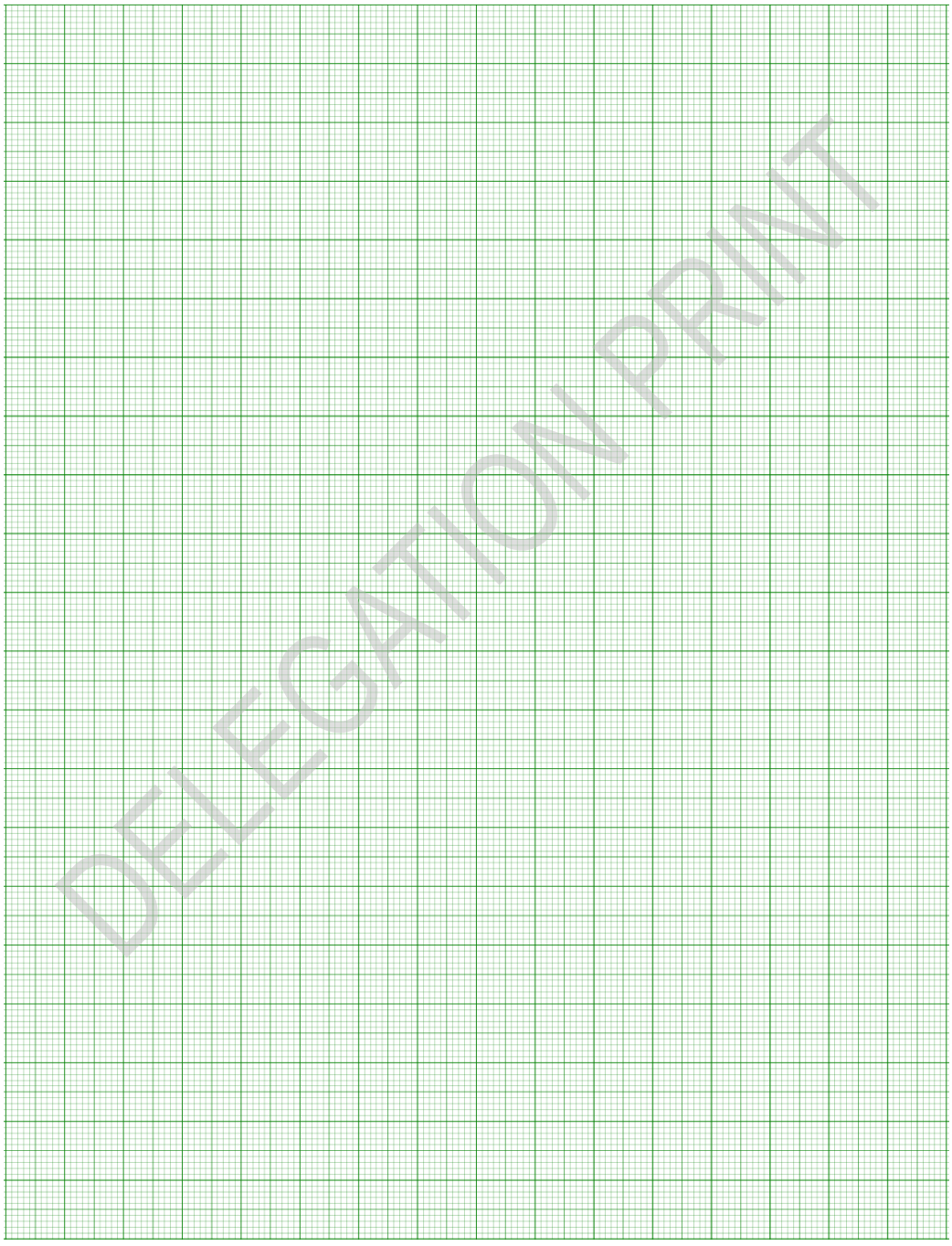
DELEGATION PRINT

$c_{Al} =$

3.4 (1.5 pt)

DELEGATION PRINT

3.4 (cont.)



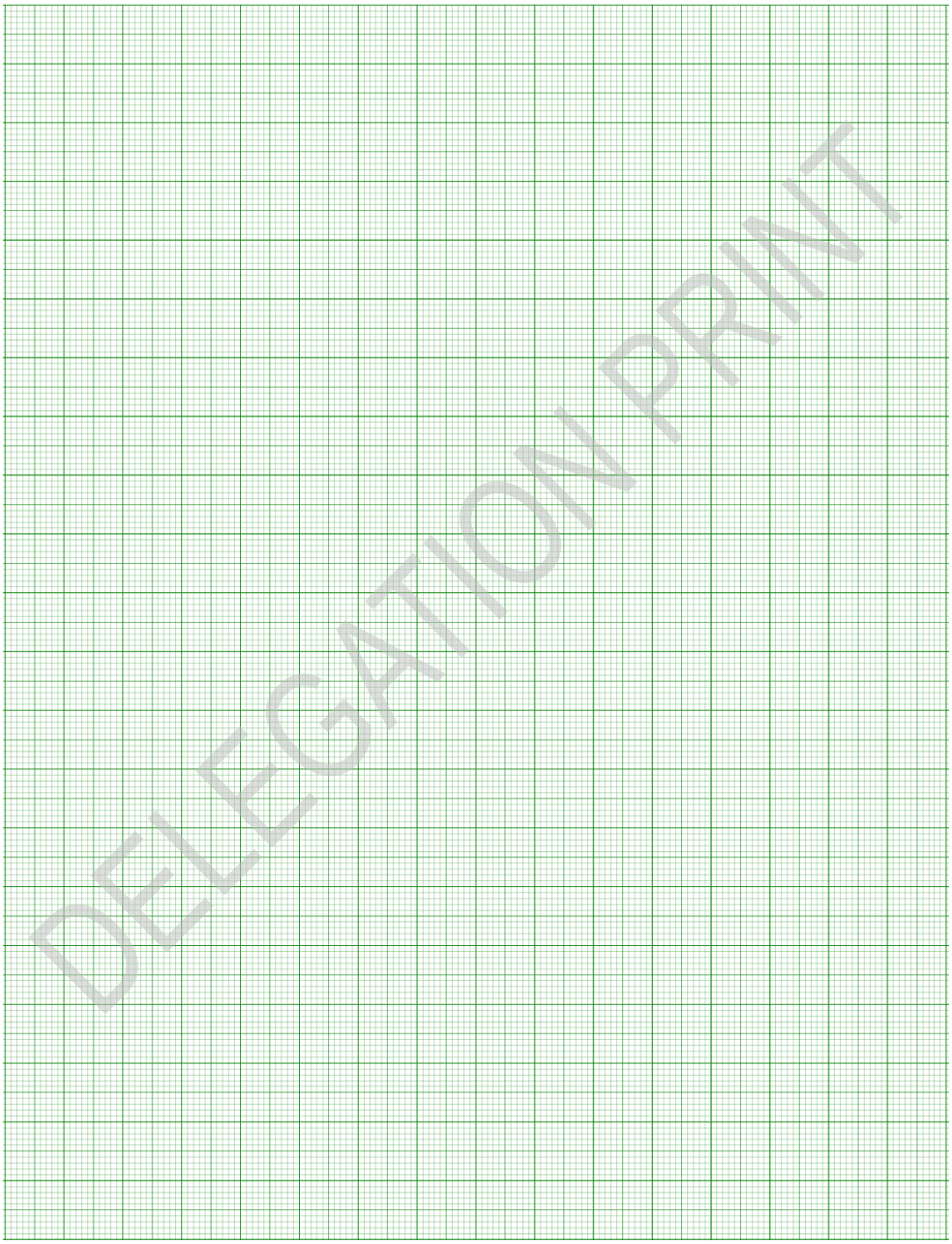
DELEGATION PRINT

$c_{SS410} =$

3.5 (1.6 pt)

DELEGATION PRINT

3.5 (cont.)



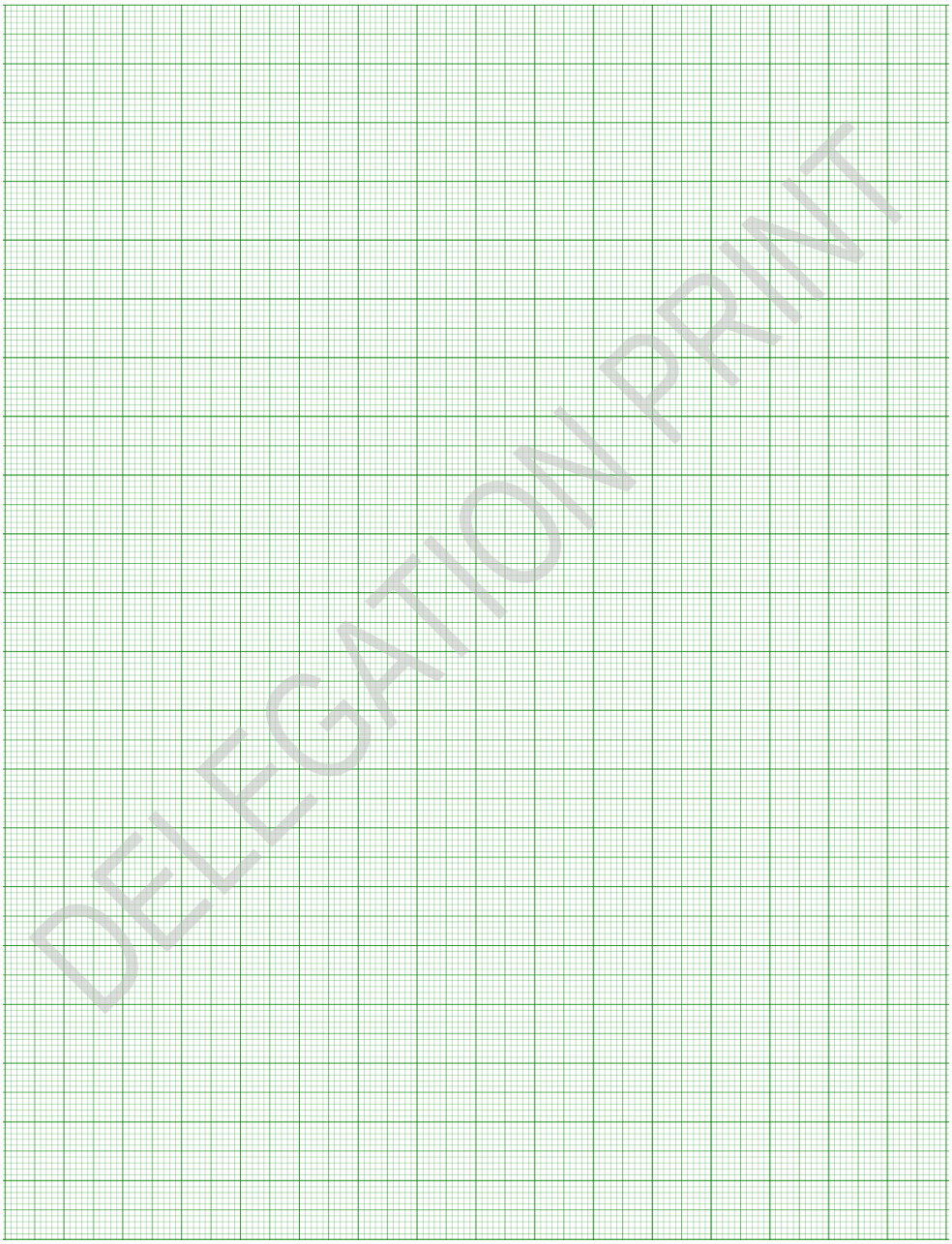
DELEGATION PRINT

$R_{LOAD} =$

3.6 (1.5 pt)

DELEGATION PRINT

3.6 (cont.)



DELEGATION PRINT

$R_{LOAD} =$

3.7 (0.1 pt)
☐ (a) ☐ (b)

3.8 (0.1 pt)
☐ (a) ☐ (b) ☐ (c) ☐ (d) ☐ (e)

3.9 (0.4 pt)

$\eta_{Al} =$ $\eta_{SS410} =$

DELEGATION PRINT